PRANAY V

VLSI Design & Verification | RTL Design (Verilog HDL) | FPGA & ASIC Synthesis | Embedded Systems

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PROFESSIONAL SUMMARY

Final-year ECE student (**B.E., VLSI Honors, CGPA 9.06/10**) who designs and verifies digital hardware in **Verilog HDL**, from specification and RTL through simulation, lint, synthesis and FPGA validation. Built a **set-associative cache controller** (218/218 regression tests), an **ARINC 429 controller** (271/271 tests with fault injection) and an **FSM design closed at 200 MHz** in Cadence Genus, with hardware experience from four internships. Seeking an entry-level role in **RTL design, design verification or FPGA prototyping**.

TECHNICAL SKILLS

RTL Design: Verilog HDL, synchronous design, FSMs, datapath and control logic, cache and memory hierarchy, input synchronization
Verification: Self-checking testbenches, reference-model (scoreboard) checking, trace-driven regressions, fault injection, lint
EDA & Implementation: Cadence Genus, Quartus Prime, ModelSim, Icarus Verilog, Verilator, Yosys; synthesis, timing and PPA analysis
Programming & Scripting: Python (reference models, regression and report scripts), Tcl, C, C++, Embedded C, Linux, Git
Interfaces & Embedded: I2C, SPI, UART, CAN / CAN FD, LIN, ARINC 429, ESP32, sensor integration, PCB design

VLSI PROJECTS

Lightweight Adaptive Reconfigurable Cache Controller (32-bit) *GitHub*

Verilog HDL, Python, Icarus Verilog, Verilator, Yosys + ABC, Nangate 45 nm

- Designed a parameterized **4 KB (1 to 16 KB) 2/4-way write-through data cache** with **Random, LRU and Pseudo-LRU replacement**, an 8-word refill FSM and **10 saturating performance counters**.
- Added a runtime policy selector using **set dueling** (leader sets, windowed miss counters, hysteresis) built from counters and an FSM.
- Verified with **218/218 passing regression tests**, comparing **about 3.7 million requests** one by one against an independent **Python reference model**; Verilator lint-clean across 6 configurations.
- Synthesized on **Nangate 45 nm**: the adaptive logic added only **0.43% area** and cut stall cycles by up to **38.7% versus LRU**.

ARINC 429 Health-Aware Communication Controller (Final-Year Project) *GitHub*

Verilog HDL, Icarus Verilog, Python

- Built **TX, RX and controller RTL** for ARINC 429 (32-bit words, odd parity, 100 kbps) with **2-FF synchronizers** and mid-bit sampling.
- Wrote self-checking testbenches and a **fault-injection campaign**: **271/271 tests pass** with **100% single-bit parity error detection**; the fault data feeds the next phase, ML-based link-health classification.

Hand Gesture to Speech Conversion on FPGA

Verilog HDL, Cadence Genus, Quartus Prime, ModelSim, Altera FPGA

- Designed real-time **RTL with an FSM-based control unit** converting flex-sensor and MPU6050 data to speech; verified in **ModelSim**.
- Synthesized in **Cadence Genus** at **200 MHz** (**critical path under 2 ns**, **power under 25 mW**); validated on an **Altera FPGA**.

WORK EXPERIENCE

Embedded Engineer Intern, Tivor, Chennai *Sep 2026 - Present*

- Building end-of-line test software for automotive electronics (**CAN / CAN FD, LIN**, multi-channel I/O, SCPI instruments, GigE vision) with a **hardware abstraction layer and runtime driver binding**; proof of concept running on a live production line.

Embedded and Software Engineer Intern, Rela Hospitals, Chennai *Feb 2026 - Apr 2026*

- Built an ESP32-based, non-invasive blood group prediction system (**94% accuracy in under 5 s**), tested against clinical requirements.

Industrial IoT Project Intern, Ashok Leyland, Chennai *Nov 2025 - Apr 2026*

- Built ESP32-to-MQTT battery and GPS monitoring for **30+ forklifts**, cutting manual inspection **60%** and unplanned downtime **40%**.

RAG and Application Development Intern, Playful Python, Chennai *Dec 2024 - Jan 2025*

- Built a production RAG application over **1,000+ documents**, improving retrieval precision by **35%**.

PATENTS & ACHIEVEMENTS

- **Two published Indian patent applications** (June 2026): non-invasive blood group prediction from fingerprints (IN202641051215 A1) and IoT-based adaptive irrigation (IN202641051218 A1).
- Winner or top performer at **15+ national-level hackathons**; **Smart India Hackathon 2025 national finalist**; Top 10 at PEC Hacks 4.0.
- **Joint Secretary**, ECE Association, SVCE: coordinated the department's technical events and student activities.

EDUCATION

Sri Venkateswara College of Engineering, Sriperumbudur, Chennai *2023 - 2027*

B.E. Electronics and Communication Engineering, **VLSI Honors** | **CGPA: 9.06 / 10**

CERTIFICATIONS

VLSI Digital Design: Chip Design and Verilog Programming (Infosys Springboard) | Application-Driven Electronics & Embedded Technology with IoT (IITM Pravartak) | Oracle Agentic AI Foundations (Oracle University) | Concepts of Cybersecurity (Microsoft) | Getting Started with Docker (Simplilearn) | Embedded Systems & Machine Learning (RAIC, ECE Association)